Workshop Pole Barn: Spec Quick Reference

Print this page and take it to every vendor call · 32 × 48 ft · 16 ft eave · gambrel · Prairie, Idaho · planned for 2028

> **Verify with the county and an Idaho engineer before ordering.** The snow, wind, frost, and seismic values below are Elmore County's published ones (site baselines); the rest are design requirements to confirm with the engineer.

Design Loads (verify with the county)

Parameter	Value	Notes
Ground snow load (Pg)	**120 psf**	The county's snow-load map puts Prairie in the 120 psf zone and requires an Idaho engineer's design there. The county ties the rule to buildings above about 5,000 ft; the lot is at about 4,750 ft, so confirm.
Flat-roof snow (approx.)	≈ 84 psf	0.7 × Pg. The truss designer calculates the real figure, with the reductions for the slopes.
Snow drift at the pitch break	Per engineering	Gambrels collect drift at the knuckle.
Basic wind speed (V)	**115 mph**	The county's residential figure (IRC); commercial uses 95 to 120 mph (IBC). ASCE 7, Risk Category II.
Wind exposure	**C**	Open ranch land.
Seismic	County zone C	Prairie is south of Featherville. The engineer sets the design category.
Frost	County 30 in zone	The county asks for footings "30 in below the frost line," which is **60 in** below grade. Footings are specified to 60 in below the pad surface. Confirm the wording.
Loft live load	40 psf minimum (60 is better)	Spec to the truss plant.
Height (county)	Mid-gable 23′-1″ above the pad; ridge 29′-1″	An accessory building may be no taller than the house or 24 ft, whichever is greater, measured to the mid-point of the gable: under by 11″ from the pad surface (about 24.5 ft if the county averages the gambrel's whole gable). The zone limit is 35 ft. See County rules on the overview page.

Pad and Foundation

Item	Spec
The pad	About 56 × 42 ft, level, built of imported fill. The fill is almost 2 ft over the original ground on the north edge and 6 in to 1 ft on the south edge (owner, October 2026). The building is centered on it with 4 ft of pad at each end and 5 ft along the long sides.
Fill	Ask the engineer whether it was compacted in lifts. If that is not documented, compact and test it before the slab, or deepen the gravel base. Check the north edge, where the fill is thickest and the margin is only 5 ft.
Pier holes	**30 in Ø × 60 in deep** minimum, measured from the pad surface. They pass through the fill and bear on undisturbed native soil, or on fill the engineer accepts. About 3 ft of native soil under the north row, 4 to 4½ ft under the south row.
Footing pad	30 in Ø × 12 in thick, 4,000 psi concrete, with an 18 in Ø tube pier above it, to the bracket.
Posts and brackets	3-ply 2×8 laminated, **wet-set on concrete piers** (Perma-Column or Sturdi-Wall Plus). 19 in all: 14 on the sidewalls, 3 on the west gable (between the corners), and 2 door jambs on the east gable.
Post spacing	**8 ft OC maximum** on the sidewalls. Layout 8-8-8-8-8-8 for the 48 ft length, so the 16 ft girts land on posts; the west gable posts are 8 ft apart.
Slab	**5 in**, 4,000 psi, #4 rebar at 16 in OC each way, with control joints at about 11 ft. It slopes toward the east door; the amount is open, because ¼ in/ft would drop 12 in over this length (see the overview's open questions).
Under the slab	4 in compacted gravel, then a 10 mil vapor barrier, then 4 in rigid foam, on the compacted pad fill. The stack puts the floor about 1′-1″ above the pad surface.

Structure

Item	Spec
Girts	2×6 SPF #2, **bookshelf at 24 in OC**
Trusses	Pre-engineered gambrel attic truss, 32 ft span, **2 ft OC**, PE stamped: 25 in all (23 common and 2 gable-end). The loft stair opening, the loft doors, and the hoist beam go on the truss drawings.
Truss heel height	12 in minimum (for insulation depth)
Truss carriers	(3) 1¾″ × 9¼″ LVL, side-bolted to each post with (3) ½ in galvanized through-bolts (Detail 7)
Purlins	2×6 SPF, 16–24 in OC per the truss design
Splash board	2×10 PT
Door header	(3) 2×12 LVL or an engineered steel beam (PE-sized)
Hurricane ties	H2.5A or stronger, **2 per truss bearing** (100 in all)

Envelope

Wall assembly (outside → inside)

Layer	Spec
1. Siding	26 ga steel, vertical panels, **barn red**
2. Rainscreen	**1×4 PT furring at 24 in OC, vertical** (critical: do not skip)
3. WRB	Tyvek CommercialWrap, taped seams
4. Sheathing	**½ in OSB**, APA Exposure 1, 8d ring-shank at 6 in on the edges and 12 in in the field (4 in at corners)
5. Framing	2×6 girts, bookshelf, at 24 in OC
6. Posts	3-ply laminated at 8 ft OC

Roof assembly (outside → inside)

Layer	Spec
1. Roofing	26 ga steel with **factory Dripstop**, **beige**
2. Underlayment	Synthetic (Titanium UDL30 or equal) over the entire roof
3. Ice and water shield	FULL coverage on the lower (steep) slope; at the pitch break and around penetrations on the upper slope
4. Deck	**⅝ in OSB** with H-clips between the purlins
5. Purlins	2×6 per the truss design
6. Trusses	Gambrel attic, 32 ft at 2 ft OC
Snow guards	On the lower slope over the walk door, and over any other door under a roof edge. The east gable has no roof slope above it, so it needs none. The engineer or the steel supplier lays out the rows for the snow zone.

Openings

Opening	Spec	Location
Overhead door	14′ × 14′ insulated, R-12 or better, **wind rated for 115 mph**	Centered on the EAST gable
Loft double doors	5′ × 6′ insulated double doors (French or barn-door style), upper lites optional	EAST gable, **centered above the main door at loft-floor level**
Landing deck	14′ wide × 10′ deep (matches the main door), free-standing: 6×6 PT posts to grade piers, doubled 2×10 PT beams east-west, 2×10 PT joists at 12 in OC north-south, 5/4 or 2×6 PT decking, 42″ rail, diagonal knee braces	Free-standing off the east gable at the loft doors
Walk door	3′ × 7′ insulated steel, pre-hung	SOUTH sidewall, near the east end
Sidewall windows	4′ × 3′ (wide × tall) insulated vinyl sliders, low-E glass, **sill 5′ above the floor**, DP-30 or better (ask the engineer for the rating at 115 mph). Nail-fin frames with jamb extensions for the wall, which is about 6¾″ thick.	NORTH wall: 3, centered in bays 2, 4, and 6, counting from the west end. SOUTH wall: 3, in bays 3, 4, and 5, clear of the stair and the walk door.
Loft window	3′ × 4′ (wide × tall) insulated vinyl single-hung, low-E glass, **sill 3′ above the loft floor**	WEST gable, centered, at the loft

Window pre-planning

- Each window is centered **between two posts**, so no post is moved. At each opening cut the girts, set a **doubled 2×6 header** on jack framing, and add a 2×6 sill and blocking for the J-trim.
- Flash every window the same way: **sill pan first, then the jamb tape, then the head tape**, lapped under the housewrap above and over it below. The rainscreen cavity stays open behind the trim, and the steel is finished around each window with J-trim and a head flashing.

Loft door and deck pre-planning

- Tell the truss plant the **loft door rough opening** on the east gable: about 5′-6″ wide × 6′-6″ tall, centered, bottom at the loft floor.
- Header above the loft doors: (2) 2×10 LVL, or as the engineer says.
- **Hoist beam included:** a cantilevered steel I-beam (or doubled LVL) projecting about 3 ft beyond the loft doors, sized by the engineer for a 500–1,000 lb working load. Anchor it to the truss top chord and to blocking inside the loft, with an eye plate or trolley track for the chain fall.
- Deck piers: 4 concrete piers below frost (60 in), 18 in Ø tubes with a belled footing. Anchor the 6×6 posts with post bases. The posts stand on the door-jamb lines, so they do not narrow the door.
- The deck is **free-standing** (no ledger on the building), which protects the rainscreen and the siding from leaks.
- The four posts need lateral bracing (14 ft wide and 16 ft tall is tall and narrow): diagonal knee braces or X-bracing per the engineer.
- Decking: PT, composite, or Trex, the owner's choice, with stainless or hot-dip galvanized fasteners.
- **No awning** over the deck. **No exterior stairs** to grade: the deck is reached from the loft only.

Loft Stairs

Item	Spec
Layout	L-shape with a 4′ × 4′ landing at the **southwest corner**
First flight	SOUTH sidewall, east to west, about 12 ft of run, rising 8 ft
Second flight	WEST gable wall, south to north, about 12 ft of run, rising 8 ft
Width	48 in clear minimum
Rise and tread	7 in rise and 11 in tread (target): 14 risers and 13 treads in each flight
Handrails	Both sides, full length, 34–38 in
Guard at the loft	42 in minimum around the loft opening
Stringers	2×12 PT or LVL
Opening in the trusses	About 4 × 8 ft at the west wall, off the southwest landing: tell the truss plant when ordering

Mechanical and Electrical

Item	Spec
Service	200 A sub-panel. **The feed source is not settled:** the pump house (inverters and batteries) is about 29 ft south of the south wall, the solar container about 33 ft south, and the planned house about 54 ft to the southeast. Check that the source can supply the loads below.
Feeder conduit	2 in PVC Schedule 40 underground, before backfill, entering at the panel
Welder	240 V / 50 A dedicated
Air compressor	240 V / 30 A dedicated
RV plug	30 A TT-30R, near the RV bay
Mini-split	240 V / 15–30 A per the condenser's nameplate
Wood stove blower	120 V / 15 A near the stove
Lighting	LED high-bay, 4000–5000 K, 15,000+ lm each, on several circuits
Heat	Wood stove (primary) plus a mini-split heat pump
Cooling	The mini-split (same unit)
Plumbing and drains	NONE, and no floor drain

Colors

Surface	Color
Roof steel	Beige
Wall steel	Barn red (dark, deep)
Trim	Beige (matches the roof)
Wainscot	None

What NOT to do

- ❌ Steel screwed straight to the OSB (no rainscreen): a sure condensation failure
- ❌ 15 or 30 lb felt behind the wall steel: it tears and traps moisture
- ❌ Poly vapor barrier outside the OSB: it traps moisture and rots the sheathing
- ❌ Foil-faced bubble wrap as the WRB: it is not a real WRB
- ❌ Ice and water shield on the walls: a vapor trap, the wrong tool
- ❌ Wood posts buried in the dirt at this snow load: use brackets on piers
- ❌ Skipping the H2.5A hurricane ties: 115 mph wind needs them
- ❌ Skipping the rebar in the slab: 5 in of concrete under stored loads will crack
- ❌ Pouring the slab or setting footings on uncompacted fill without the engineer's say-so

Mountain Prairie Acres · Workshop pole barn spec quick reference · updated October 23, 2026 · counts and sizes are filled in from `docs/workshop_model.js`.